

LAMBA

A digital twin for a car

Android app, FastAPI backend, PostgreSQL timeline, and AI-assisted chat for structured vehicle records.

PROBLEM

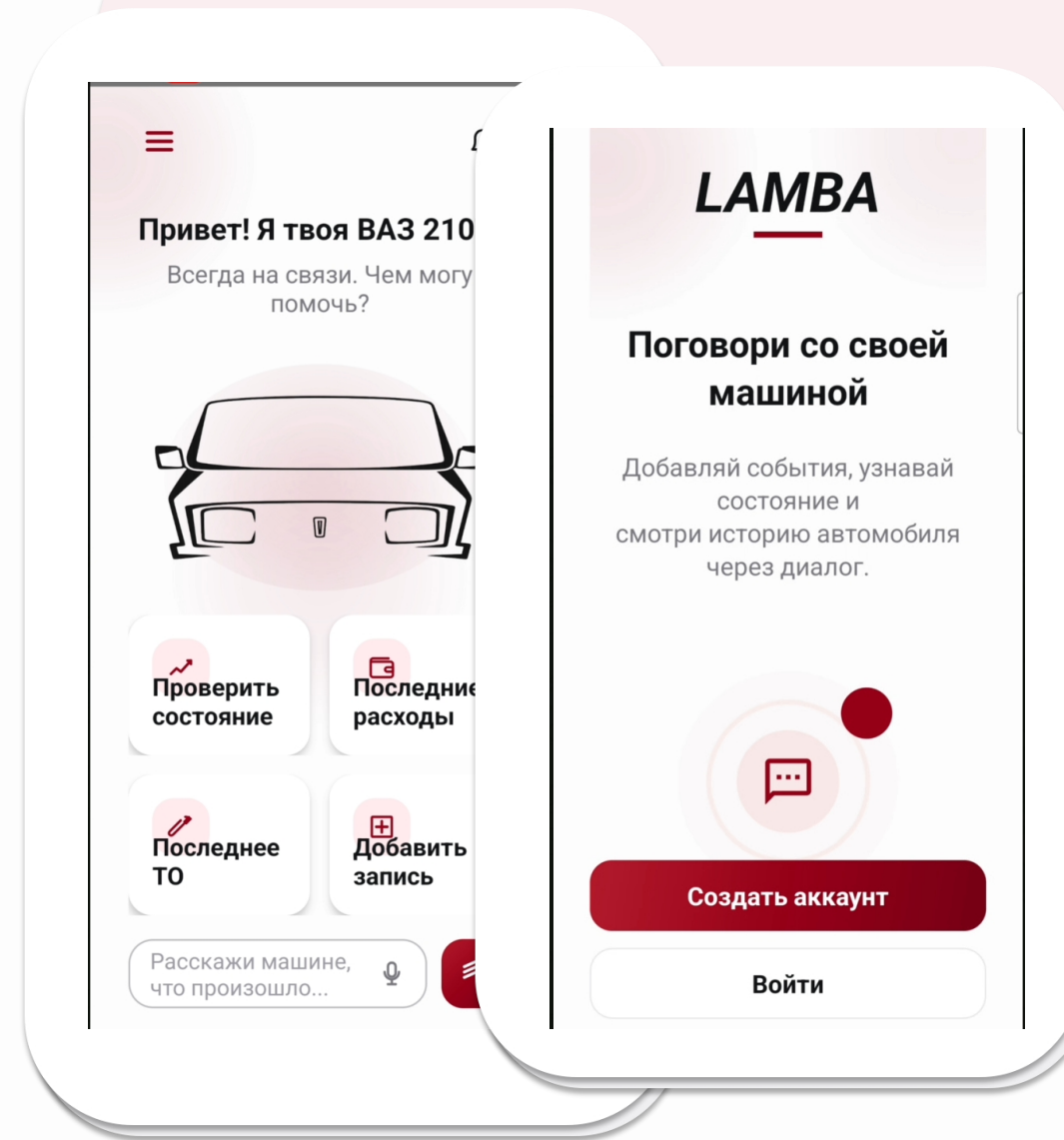
vehicle data loss

SOLUTION

AI-twin for the car

END-USERS

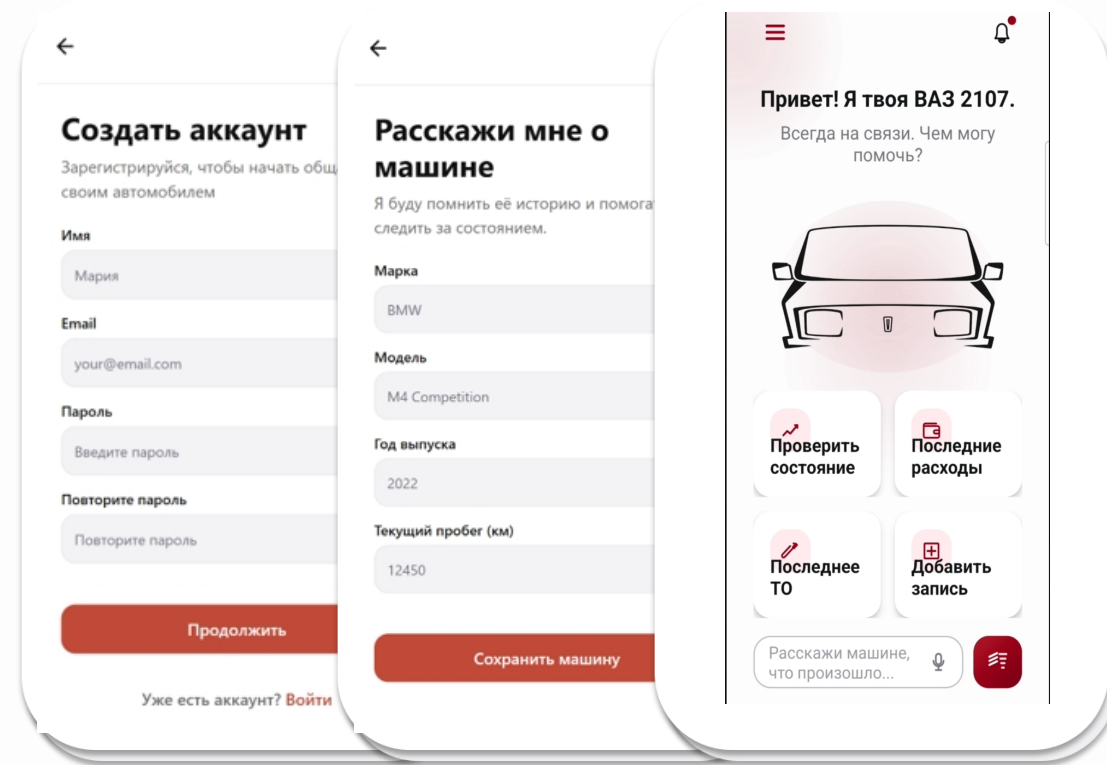
car owners



Show the product flow first.

Video demo

here will be the link to the video demo



register

vehicle

chat

1

Register

Create account or log in.

2

Vehicle

Add your car profile.

3

Chat

Send natural message.

4

Timeline

Review structured record.

TEAM

Responsibilities map directly to the increment.

Each role owns a visible part of the MVP evidence.

VT

Vasilisa Tumakina

Team lead / backend

Planning, backend coordination, release evidence, AI integration, chat parser

EI

Elizaveta Ionina

Frontend developer

Onboarding, login/register integration

MC

Maria Chizhikova

Frontend developer, UI/UX

Vehicle/profile screens and UI flow support

AK

Amira Khaliullova

Backend developer

Auth, vehicle endpoints, validation/tests

IG

Ivan Grishaev

AI engineer

Chat parser, AI integration, event extraction

REQUIREMENTS

MVP scope is clear, quality gates stay visible.

The deck separates must-have delivery from later expansion.

Must have

- Registration
- Add vehicle
- Chat messages
- Records from chat
- Timeline
- Assistant
- Basic statistics

Should / could

- Maintenance recommendations
- Notifications
- Voice messages
- Repair receipts

Out of MVP

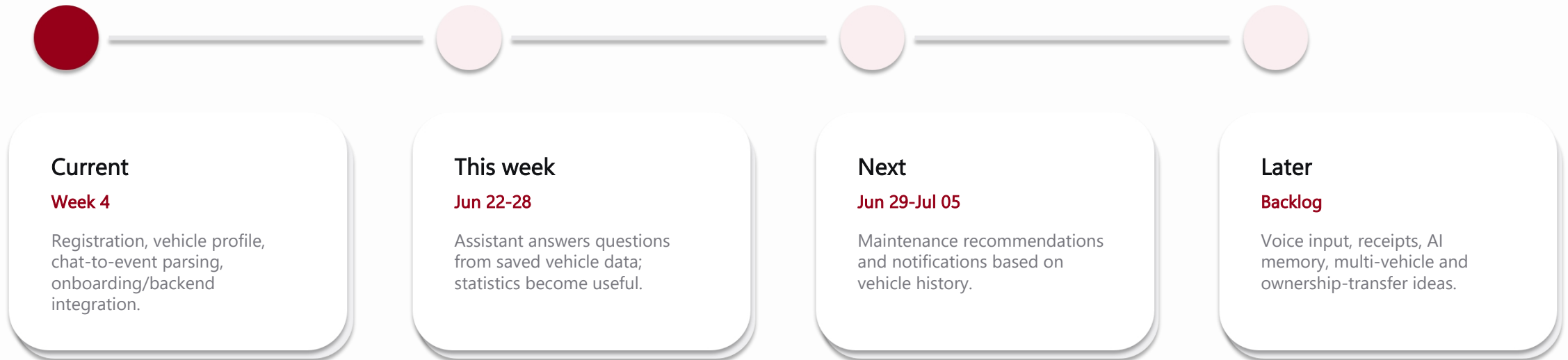
- Multi-vehicle management
- Ownership transfer
- OBD-II integration

Quality

- No plain-text passwords
- User-specific backend data
- Automated tests
- Review, changelog, docs

Stabilize the core flow before expanding AI behavior.

The next work is sequenced around user value and delivery risk.



Customer feedback is already shaping scope.

The team is keeping MVP decisions small, usable, and demoable.

Already usable

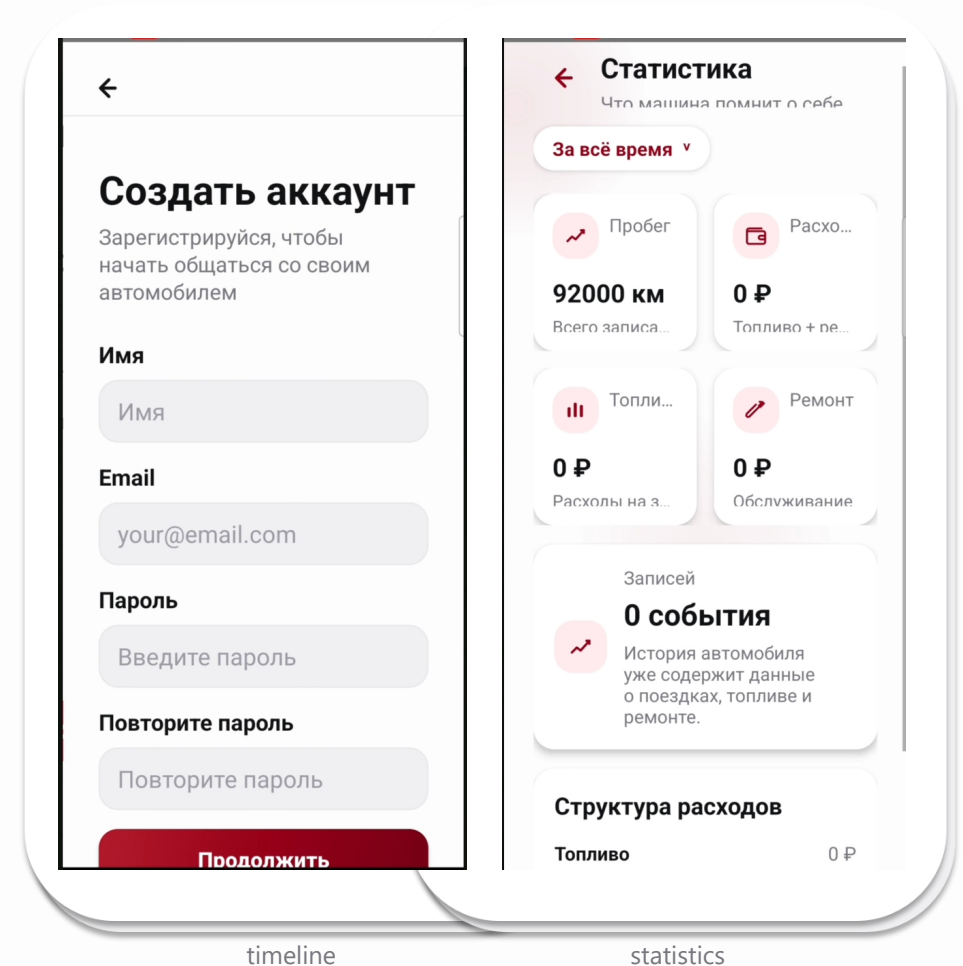
Deployed backend and Swagger · Repeatable smoke-check flow · Android screens for core flow

Useful feedback

Login + password is enough · Manual brand/model input is acceptable · One vehicle per user is acceptable

Current challenges

Password security hardening · User-specific data verification · Keep advanced AI outside MVP



Repository and deployed API

GitHub repository

<https://github.com/LAMBA-23/LAMBA>



Deployed Swagger UI

<http://10.93.26.193:8000/docs>

